Identifying Crop Change from Hail Damage Using Python and HLS Data

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Hail Scars, Importance, and Stakeholders

- Much of the Great Plains and Midwest grow crops during peak hailstorm months (late-Spring into Summer). These storms cause thousands of dollars in damages to crops, and can often be seen from satellite images.
- Farmers and crop insurance adjusters are the main stakeholders from a hail scar database. A database would provide better understanding of typical time of year, frequency, and size, and could help farmers plan accordingly.
- Current method of identification of hail scars is manual. With hundreds occurring in a year, it is a monotonous process.
- The goal is to create an automation to detect hail scars using land change products like NDVI.

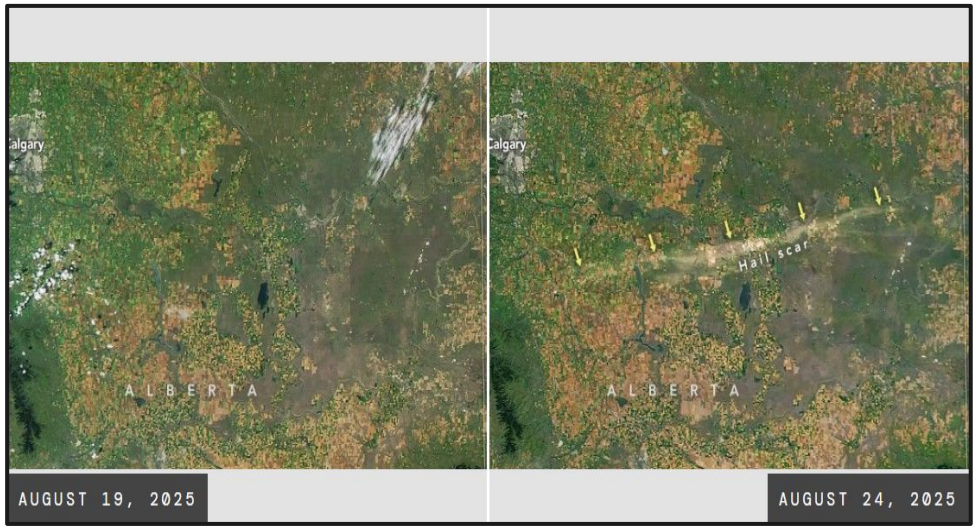


Fig. 1 - Hail scar in Alberta, Canada seen from NASA's MODIS satellites in August 2025 (NASA Earth Observatory, 2025)

Data

- HLS Sentinel-2 30m imagery obtained from NASA EarthData was used.
- To create indices like Normalized Difference Vegetation Index (NDVI), raw data was multiplied by 0.0001. Missing values, noted as -9999, were also filtered out.
- Raw data was transformed into the WGS 84 coordinate system.
- Some before images were taken a year-removed from the event as a control. Given vegetation growth times, the before images were often from the same time of year.
- I focused on three types of hail swaths:
  - Pristine - clear damage against a green, full crop background (Example used Iowa and South Dakota)
  - Semi-pristine - visible to the human eye, but nearby vegetation not as lush (Example used Montana)
  - Non-pristine - difficult to see with human eye, little to no change from nearby vegetation (Example used Montana)
- To calculate NDVI, I used Sentinel-2's Red (Band 4) and Near-Infrared (Band 8A)

Methodology

1.) **Measure Vegetation Growth Before and After Event**
Calculated NDVI of image before and after the event. NDVI measures the "greenness" of the vegetation, where healthy crop growth is a high value, near 1, and damaged crops are a lower value, near 0.

2.) **Find where vegetation depleted in after imagery**
The before image is subtracted from the after image. A large negative difference indicates possible hail damage.

1.) Measure Vegetation Growth Before and After Event

2.) Find where vegetation depleted in after imagery

3.) Clean up signals

4.) Remove small areas of change

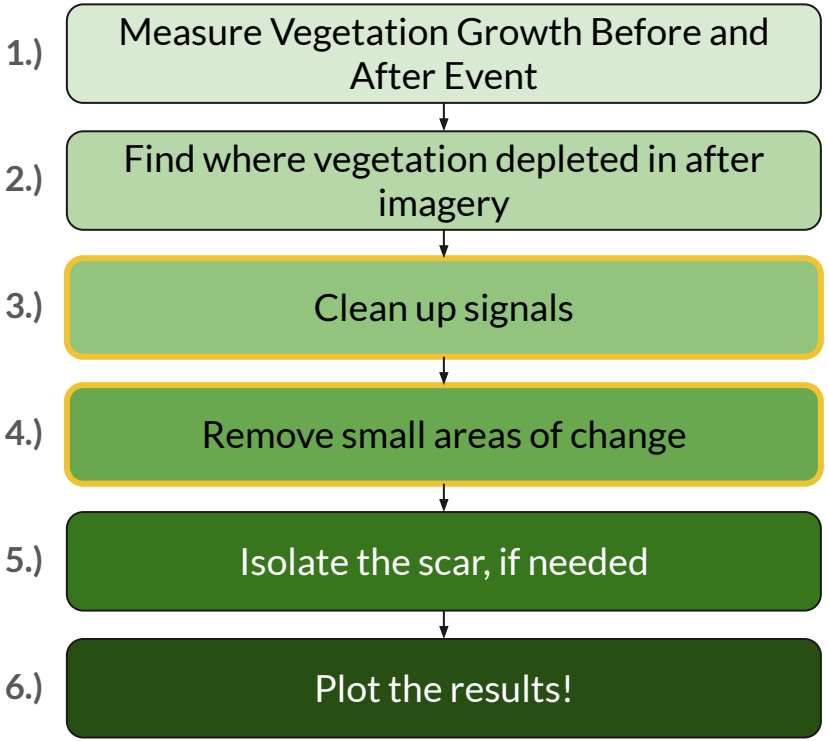
5.) Isolate the scar, if needed

6.) Plot the results!

Methodology (cont.)

3.) **Clean up signals**
A lot of the data came out noisy, and without a filter, many fields and clouds were deemed hail scars. I used an opening disk to remove isolated speckles, such as clouds or sensor feedback, and closing disk to fill in small gaps between nearby damaged areas. This way, a whole swath is indicated and not just fragments.

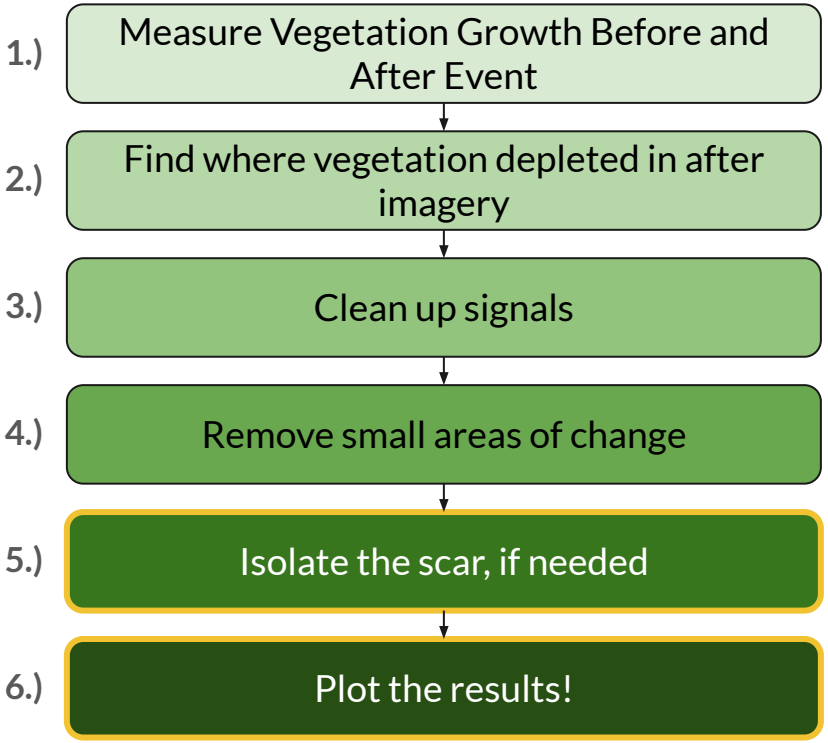
4.) **Remove small areas of change**
A minimum "blob-size" is adjustable in the function, and it filters out small agriculture field changes that might still get picked up after cleaning up the signals. It is prime crop growing season after all, so big crop changes are expected!



Methodology (cont.)

5.) **Isolate the Scar, if needed**
If the proximity filter is turned on, then the largest blob of negative change is selected, and any smaller blobs nearby are grouped in with the assumed swath. If not, then everything that passes the size filter is kept. The filter is most useful with clearer swaths.

6.) **Plot the results!**
The difference of the two images is plotted alongside what is assumed to be the scar. This is for ease of identifying with the human eyes. Provides a chance to check work!



Parameters

- **Swath Threshold Parameter**: Healthy NDVI values sit between 0.3 to 0.8. When hailstorms damage fields, the NDVI drops dramatically. A threshold closer to 0 is often too loose bc it will find values that have 0% change. But the lower the threshold, the less clear the swath might become.
- **Minimum Blob Size**: After the threshold, many areas are flagged with change. Most are small agriculture fields that have a similar change to a hail scar. This minimum size will throw away any areas that are smaller than what is asked.
- **Closing Disk Size**: Since many scars are not uniform, a threshold is needed to ensure the whole swath is picked up as a swath! The closing disk will fill in gaps between areas of high NDVI damage. The lower the value, the less uniform the function will find the scar.
- **Proximity Filter**: Despite all of this, some larger fields may still pass through the filters, especially if the scar is small or heavily fragmented. Turning on a proximity filter will automatically detect the largest blob of change, which would be the scar more than likely, and keep smaller areas of change that are nearby in the end product. Thus, fragments will be grouped as one swath, and large fields away from the scar will be filtered out.

Case 1 - Pristine in Iowa

Parameters Used:
swath_threshold: -0.18
min_blob_size: 500
dilation_disk_size: 50
prox_filter: True

This scar in particular is very pristine, where it is very clear on the NDVI difference plot.
The swath threshold is finding where there is a 18% change, which is low. Minimum blob size is high, since there are not many areas with as high values as this scar. The function can encompass most of the scar with the high minimum blob.

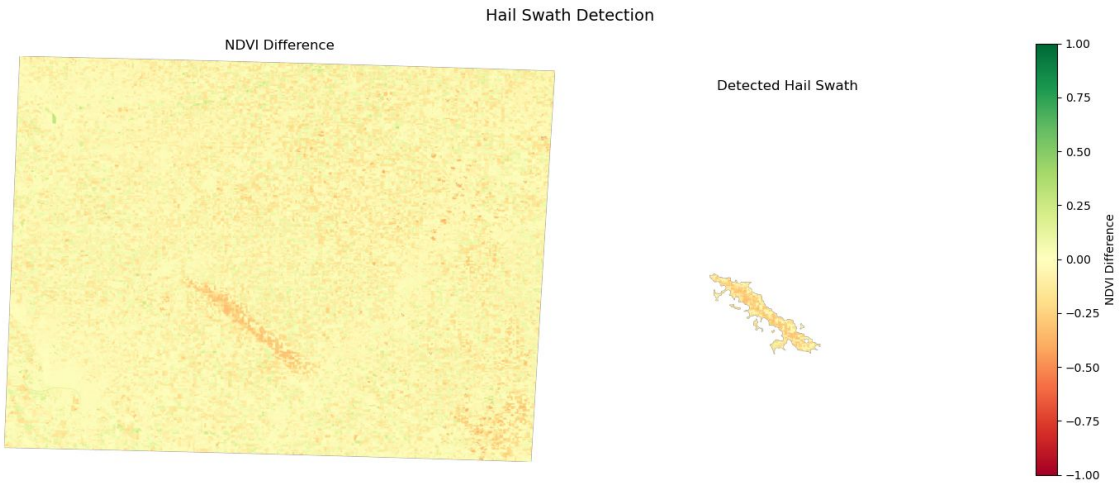


Fig. 2 - Left is NDVI Difference in Iowa, Right is the detected hail scar from Python. This was generated in Python using Sentinel-2 Data from NASA EarthData.

A high dilation disk size is not needed here, since it is a very uniform scar. The proximity filter was on due to some noise to the SE of the scar. Overall, this is the textbook definition of a pristine scar. The program had no trouble finding it.

Case 2 - Pristine in South Dakota

Parameters Used:
swath_threshold: -0.18
min_blob_size: 500
dilation_disk_size: 50
prox_filter: True

This scar is slightly less clear, as it blends in a bit more with its background. The swath threshold is finding where there is a 18% change, the same as the last pristine. Minimum blob size is high, since there are not many areas with as high values as this scar. Though if it were higher, the noise to the SE and N would be picked up, which are not part of the scar.

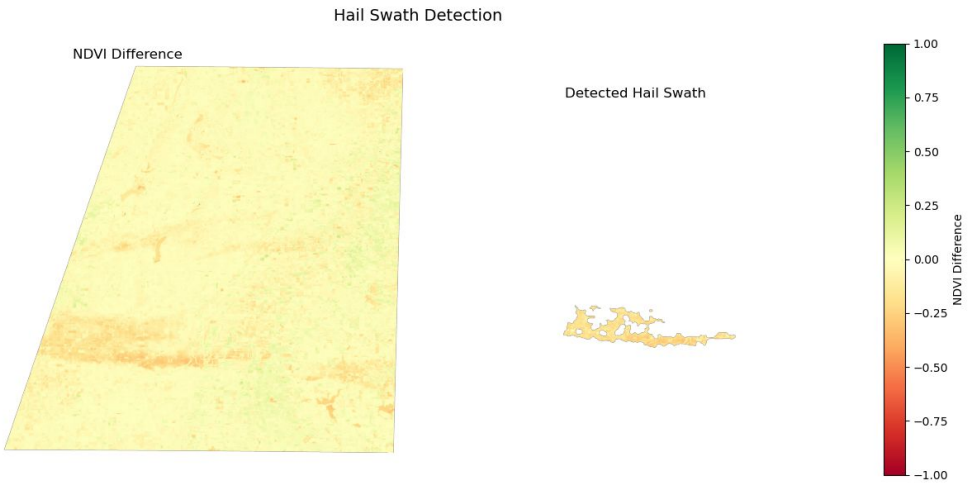


Fig. 3 - Left is NDVI Difference in South Dakota, Right is the detected hail scar from Python. This was generated in Python using Sentinel-2 Data from NASA EarthData.

A high dilation disk size is not needed here either, since overall the scar is all together. Since this scar was not as linear, the proximity filter was turned on to group in the non-linear part of the scar, which the function struggled a bit with. Though less pristine, this scar was still very clear and detected easily.

Case 3 - Semi-Pristine in Montana

Parameters Used:
swath_threshold: -0.15
min_blob_size: 200
dilation_disk_size: 300
prox_filter: True

This scar is very fragmented, though still fairly easy to see with the human eye, with multiple fields of significant NDVI change. The swath threshold is slightly higher at -0.15, due to some values of the scar being less significant of a change than the pristine examples. Minimum blob size is lower, as it was picking up non-hail damage as part of the scar at a bigger minimum blob size.

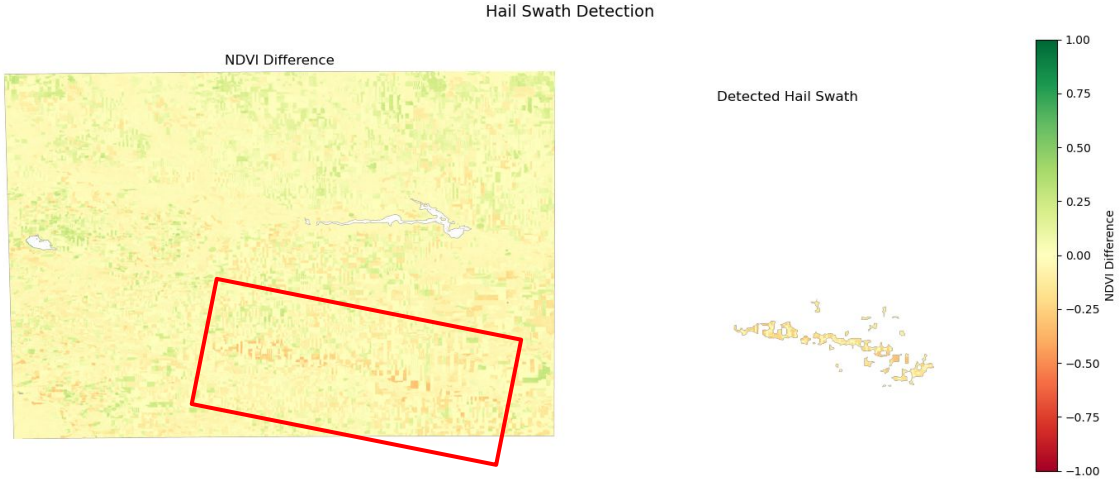


Fig. 4 - Left is NDVI Difference in Montana, Right is the detected hail scar from Python. This was generated in Python using Sentinel-2 Data from NASA EarthData.

This is a case where a high dilation disk size matters. Since the scar is so fragmented, a high dilation disk will group all the fields it deems fit into one. This is helped by the proximity filter, which gives it a more linear appearance, though some noise is still picked up to the north.

Case 4 - Stress Test in Montana

Parameters Used:
swath_threshold: -0.05
min_blob_size: 50
dilation_disk_size: 30
prox_filter: False

This scar was a complete stress-test. On visible imagery, it is hard to define, and even the difference calculation struggled. The swath threshold is very high, highlight even little changes, in hopes that the scar would pass through. Same with minimum blob size, which is much lower, trying to identify any blob of change, but was unsuccessful.

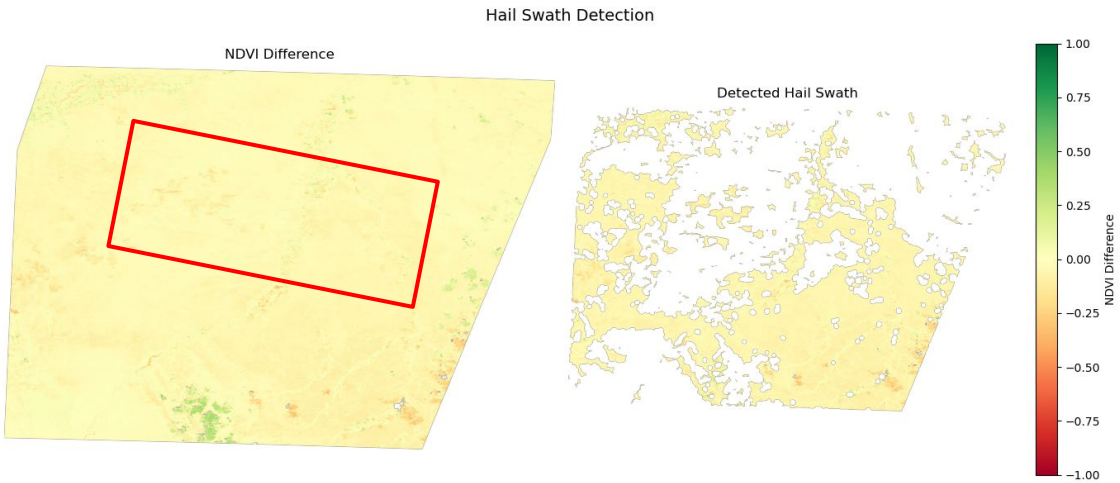


Fig. 5 - Left is NDVI Difference in Montana, Right is the detected hail scar from Python. This was generated in Python using Sentinel-2 Data from NASA EarthData.

The original scar is uniform, so trying a low dilation disk size was attempted. Without its clarity from the difference calculation, it did not seem to matter. Plus, the proximity filter being false meant any area was going to pick up, not just the biggest blob. The proximity filter on would've picked up the non-hail change to the SE.

Limitations and Assumptions

- **Agriculture and Noise**: Many of the filters and parameters were defined due to agriculture. Many fields will harvest or change on a scale similar to a hail scar, so to identify the scar rather than fields, filters were put into place.
- **Biggest Blob**: This assumes that the scar is the biggest area of change in an image. This can be false. Though not tested, some hail scars can be very small, the size of a few fields together. If an area of change is larger than the scar, that will get processed instead.
- **Cloud Cover and Imagery**: This imagery was handpicked, making sure images were not cloud covered. Imagery from something like Synthetic Aperture Radar, or SAR, would work better in a situation with cloud cover. This also hinders accuracy, as it might take weeks to find a best image. This allows the ground to recover more, leading to a weaker signal.
- **Trial and Error:** The parameters, like blob size and swath threshold, were used in trial and error and are nowhere near automated. This means a combination of parameter values might work better for each case, which is a large limitation as there is not enough time to test every possible value.

Conclusions and Potential Future Work

- This code works great with pristine swaths. However, it struggles with semi-pristine, and did not pass the stress test. The semi-pristine needed completely different parameter values, and none of work for the stress-test.
- Other products could be used that might work better. In the proposal and in testing, Normalized Burn Ratio was intended on being used, but its signals were weak. Perhaps another type of index could work, like a modified NDWI.
- Use of SAR or another function that identifies clouds and masks them out could open the floor to imagery closer to the event. In the stress-test, the imagery was weeks old, which did not help that the scar was already hard to identify.
- The goal would be to make a database of all these hail scars, and run statistics on frequency of events in a given area, most common time of year, and some math calculations like area. This would allow farmers to be more intune with hail storm season and could plan their crops around it.

References

Bell, J. R., Wisinski, E. F., Molthan, A. L., Schultz, C. J., Gilligan, E., & Sharp, K. G. (2023). Developing a Hail and Wind Damage Swath Event Database from Daily MODIS True Color Imagery and Storm Reports for Impact Analysis and Applications. *Weather and Forecasting*, *38*(9), 1575–1588. https://doi.org/10.1175/waf-d-22-0210.1

NASA Earth Observatory. (2025, August 28). *Hail Scars Alberta Farmland*. NASA Science. https://science.nasa.gov/earth/earth-observatory/hail-scars-alberta-farmland-154724/